

Repo: Gabriel-Kelvin/superrder-patch-exercise
commit 56a43e8

(I used AI (Codex) to help understand the patch. Then I wrote this in my own words. Checks Codex ran are not checks I ran.)

1. Search showed archived + wrong status tasks

Where → TaskRepository.java 1.14-16,
search_tasks.sql 1.10-14, oracle pkg 1.52-55 & 66-70

Found → no brackets around title OR description

Cause → SQL does AND before OR → title match skipped status, desc match skipped archive filter

Fix → archived=false AND (title OR desc) AND status

Why → group the OR once so results + count agree

Tests → H2: 47 visible tasks, 6 OPEN for "api".

Oracle not run (reference only)

2. Pages shifted when timestamps tied

Where → TaskRepository 1.16, H2 1.14, Oracle 1.70

Found → ORDER BY created_at only → not unique

Cause → SQL gives no order for equal timestamps

Fix → added id DESC as tie-breaker

Why → still newest first, same order every time on fixed data

Limit → concurrent inserts can still shift pages.
Ties reviewed, not tested

3. Invalid input + int overflow

Where → TaskController.java 1.27-43,

48-51

Found → unguarded Enum.valueOf, subList indexes from raw request ints

Cause → bad status throws exception, page 0 or negative = bad index, int multiply can overflow

Fix → page < 1 or pageSize not 1..100 → 400.
Unknown status → 400. Offset as long

Also → trim status, blank = no filter, Locale.ROOT for case

Why → client errors should be 4xx, math stays safe

Limit → cap limits response size but all matches still load in memory

4. Artificial delay

Where → TaskController.java old 1.41-55

(removed)

Found → `Thread.sleep` in the handler, based on query length

Cause → blank search slept 1s, thread blocked doing nothing

Fix → removed sleep + query logging, and console logging in `frontend/api.js`

Why → sleeping doesn't speed up the DB, and there's less user text in logs

Limit → no benchmark done, so no speed claims

5. Old requests overwrote new results

Where → `frontend/src/hooks/useTasks.js` l.10-34, `api.js` l.3 & 11

Found → no cleanup, catch never ended loading. Deferred-promise test finishes old request late

Cause → responses arrive out of order → old callback overwrites state, error stayed after a

later success

Fix → AbortSignal in fetch, cleanup marks effect inactive + aborts, callbacks check active, clear error at start, loading off in finally

Why → abort saves client work, the active check covers late cancels

Limit → abort doesn't stop SQL on the server

6. Filters left you on an invalid page

Where → frontend/src/App.jsx 1.24-25, 32, 38

Found → handlers changed query/status but page stayed → page 2 + 6 results = empty page

Fix → filter handlers reset page to 1, pagination disabled while loading

Why → React batches updates, so the new filter requests page

1

Tests → search + status change. Smoke test: page 2, then api + OPEN → 6 rows

7. Tests + tradeoffs

Added → TaskSearchTest.java (3 tests), App.test.jsx + useTasks.test.jsx (3 tests)

Result → all 6 tests + frontend build passed

Port → 5173 blocked (EACCES), used --port 55173. Need to check default on my machine

Left for later → no auth, H2 console on, in-memory DB, all-result pagination, leading wildcard, no debounce, npm audit, Oracle validation